Kartik Bali

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PhD candidate in 3D computer vision, multimodal LLMs (VLMs), and generative models. First author of MV-GEL (**ECCV 2026**) and GReFEM (**NeurIPS 2026**). Interested in language-driven 3D understanding, VLM post-training (SFT/GRPO), and efficient inference.

Available for research internships: Summer/Fall 2027.

Education

- **M.Sc. Computational Science & Engineering**, Technical University of Munich *Oct 2021 – June 2024*
  Grade **1.5** (German scale; 1.0 = best); focus in Machine Learning and Simulation Sciences; Deutschland Stipendium.
- **B.Eng. Mechanical Engineering**, BITS Pilani, Hyderabad *July 2015 – May 2019*
  Grade **9.63/10**; Merit Scholarship (top 2% academic performers).

Experience

- **PhD Candidate** — Helmholtz Zentrum Hereon, Hamburg *Jan 2025 – Present*
  - First author of **MV-GEL** (**ECCV 2026**): language-driven **multi-view** localization of fine-grained **geometric CAD entities** on 3D meshes; prompt-conditioned **view selection** over a **VLM** reasoning-segmentation backbone; up to **1.7**× face-level IoU and **4.5**× edge-level F1 over CLIP-based baselines.
  - First author of **GReFEM** (**NeurIPS 2026**): zero-shot **multimodal LLMs** as semantic proxies for physics-guided 3D **FEM mesh refinement**, with a trained view selector (orthoViews); without any FEM solve, closes **58%** of the gap to a reference-solve oracle.
  - **SightCAD** (under review, **ICLR 2027**): view selector and CadQuery-generating **VLM** jointly trained with **RL on execution rewards** for **CAD reconstruction under occlusion**; **+21/+17** effective-mIoU over best baselines on DeepCAD/Fusion360, ≤1.5% invalid-code rate; **mechanistic** view-token ablations show which views make which program parameters legible.
  - Ongoing — **Point2BIM** (with DFKI): real laser scans → executable **BIM programs** (IFC4) via VLMs; **SFT + GRPO** with execution-based geometric rewards; 4×H200 training, **vLLM** inference.
- **Master Thesis Student** — BMW AG, Munich *Sept 2023 – May 2024*
  - Accelerated design of 2D/3D flow geometries by **2**× using **Physics-Guided Diffusion Models** and **GANs**, integrated with an autograd-based 3D topology optimization tool. Grade **1.0**.
- **Machine Learning Werkstudent** — TUM Graphics & Visualization (Thuerey group) *May 2022 – Aug 2023*
  - Implemented **Graph Neural Networks** and **Convolutional ResNets** for chaotic PDEs; built CI/CD and tests for PhiFlow. Contributed to a **CMAME 2025** journal publication.

Publications

1. **Bali, K.** and Aydin, R., "MV-GEL: Language-Driven Multi-View Geometric Entity Localization on Meshes", *ECCV*, 2026 **(accepted)**. arXiv:2606.31533.
2. **Bali, K.**, Guru, M. K., Cyron, C. J., and Aydin, R., "GReFEM: Multimodal LLMs as Zero-Shot Semantic Assistants for Physics-Guided 3D Mesh Refinement", *NeurIPS*, 2026 **(accepted)**. arXiv:2607.08798.
3. Tacke, M., …, **Bali, K.**, et al., "Constitutive Scientific Generative Agent (CSGA): Leveraging Large Language Models for Automated Constitutive Model Discovery", *Mach. Learn. Comput. Sci. Eng.*, 2025.
4. List, B., Chen, L.-W., **Bali, K.**, Thuerey, N., "Differentiability in Unrolled Training of Neural Physics Simulators on Transient Dynamics", *Computer Methods in Applied Mechanics and Engineering*, 2025.
5. **Bali, K.**, Guru, M., Borjigin, Y., Starostina, A., Cyron, C. J., and Aydin, R., "Look Where You Can: Active View Selection for CAD Reconstruction under Occlusion", under review at *ICLR*, 2027. arXiv:2610.11954.
6. Guru, M. K., Gupta, V., **Bali, K.**, et al., "From Finite Element Analysis to Self-Optimization: FRAME and the Era of Agentic AI in Engineering", under review at *npj Computational Materials*, 2026.
7. Minder, J., …, **Bali, K.**, …, Aydin, R., West, R., "Synthetic Persona Pretraining: Alignment from Token Zero", preprint, 2026. arXiv:2608.13482.
8. **Bali, K.** and Supradeepan, K., "Numerical Analysis of Flow Past an Oscillating Circular Cylinder using Arbitrary Lagrangian Eulerian Consistent Flux Reconstruction Scheme", 2019. arXiv:1911.03415.

Skills

- **ML/AI:** Vision-Language Models (VLMs), Multimodal LLMs, LLM Fine-tuning, RL Post-training (RLHF/GRPO), Diffusion Models, GANs, GNNs
- **Computer Vision:** 3D Reconstruction, Multi-View Geometry, Referring Segmentation, Detection, Point Clouds, Meshes
- **Tools:** Python, C++, PyTorch, HuggingFace, vLLM, Distributed Multi-GPU Training, JAX, CadQuery, BlenderProc, PyVista, Git, Linux
- **Domain:** Scientific ML, Differentiable Simulation, PDEs, Generative Design **Languages:** English (C1), German (B1)